

Exploratory Data Analysis (EDA) Summary Report Template

1. Introduction

This document provides an exploratory analysis of Geldium's dataset with the goal of assessing data quality, discovering meaningful insights, and identifying the factors that may contribute to credit default risk. The analysis is intended to support effective predictive modeling and improve risk assessment accuracy.

2. Dataset Overview

The dataset consists of 500 customer records from Geldium and contains key variables associated with credit delinquency. It includes both numerical and categorical features such as income, credit utilization, missed payments, and debt-to-income ratio.

Key dataset attributes:

- Number of records: 500
- Key variables: Age, Income, Credit Score, Credit Utilization, Missed Payments, Debt-to-Income Ratio
- Data types:
 - Categorical variables: Employment Status, Credit Card Type
 - Numerical variables: Income, Loan Balance

3. Missing Data Analysis

Several important variables contain missing values, particularly the income and loan balance fields. If not properly handled, these missing entries may negatively impact the performance and reliability of predictive models.

Key missing data findings:

- Variables with missing values:
 - Income: 50 missing records
 - Loan Balance: 30 missing records
- Missing data treatment:
 - Impute missing numerical values using the median

- Utilize AI-generated synthetic data where suitable for Loan Balance

4. Key Findings and Risk Indicators

The analysis revealed a strong relationship between high credit utilization and delinquency, along with an increased risk linked to repeated missed payments.

Key findings:

- Customers utilizing more than 50% of their available credit are generally at higher risk.
- Customers with 3 or more missed payments within a six-month period are more likely to default.
- Certain irregularities were identified, including high-income individuals with unexpectedly low credit scores, which require further investigation.

5. AI & GenAI Usage

Generative AI tools were used to assist in identifying patterns, detecting missing values, and analysing risk-related factors. The AI-generated findings were then validated against standard financial risk assessment measures.

Example AI Prompts:

- “Summarize data patterns and identify missing values.”
- “Evaluate default risk based on credit utilization and payment history.”

6. Conclusion & Next Steps

The exploratory data analysis provided valuable insights into Geldium’s dataset by identifying missing data issues, uncovering behavioural trends related to credit risk, and highlighting unusual cases that may require additional review.

Main Findings:

- Missing values in income and loan balance fields may affect analytical outcomes.
- High credit utilization and repeated missed payments are strong indicators of delinquency risk.
- Unusual cases involving high-income customers with low credit scores should be examined further.

Recommendations:

- Apply appropriate imputation techniques for missing income and loan balance values to reduce bias.
- Validate whether the identified risk indicators remain consistent across different customer segments.
- Review abnormal data entries to ensure data accuracy and identify potential financial concerns.

These actions will help Geldium strengthen its risk assessment framework and improve data quality for future predictive modelling efforts.